

Algebra

$$\sin n\pi = 0$$

$$1 - \cos n\pi = 2 \text{ for odd } n$$

$$\arctan\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}, \arctan(\sqrt{3}) = \frac{\pi}{3}$$

$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$$

$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$$

$$\sin x \sin y = \frac{1}{2}[\cos(x - y) - \cos(x + y)]$$

$$\cos x \cos y = \frac{1}{2}[\cos(x - y) + \cos(x + y)]$$

$$\sin x \cos y = \frac{1}{2}[\sin(x - y) + \sin(x + y)]$$

$$\sin a \pm \sin b = 2 \sin \frac{a \pm b}{2} \cos \frac{a \mp b}{2}$$

$$\cos a + \cos b = 2 \cos \frac{a+b}{2} \cos \frac{a-b}{2}$$

$$\cos a - \cos b = -2 \sin \frac{a+b}{2} \sin \frac{a-b}{2}$$

$$Cc(\omega_0 t + \theta) = Cc(\theta)c(\omega_0 t) - Cs(\theta)s(\omega_0 t)$$

$$Cs(\omega_0 t + \theta) = Cs(\theta)c(\omega_0 t) + Cc(\theta)s(\omega_0 t)$$

$$\theta = \tan^{-1}\left(-\frac{b}{a}\right), \pm\pi \text{ when } a < 0$$

$$\sin t = \cos\left(t - \frac{\pi}{2}\right)$$

$$-\cos t = \sin\left(t - \frac{\pi}{2}\right)$$

$$\cos x = \frac{1}{2}[e^{jx} + e^{-jx}]$$

$$\sin x = \frac{1}{2j}[e^{jx} - e^{-jx}]$$

$$e^{j\omega t} = \cos(\omega t) + j \sin(\omega t)$$

$$z^* = a - jb = re^{-j\theta}$$

$$u^*v^* = (uv)^*$$

$$\angle z = \tan^{-1}\left(\frac{b}{a}\right), \pm\pi \text{ in Q2 and Q3}$$

$$z^{\frac{1}{n}} = r^{\frac{1}{n}} e^{j\frac{\theta+2\pi m}{n}}$$

Integrals

$$\int \cos^2 at \, dt = \frac{t}{2} + \frac{\sin 2at}{4a}$$

$$\int t \cos at \, dt = \frac{1}{a^2}(\cos at + at \sin at)$$

$$\int t \sin at \, dt = \frac{1}{a^2}(\sin at - at \cos at)$$

$$\int t^2 c \, at \, dt = \frac{1}{a^3}(2atc \, at - 2s \, at + a^2 t^2 s \, at)$$

$$\int t^2 s \, at \, dt = \frac{1}{a^3}(2ats \, at + 2c \, at - a^2 t^2 c \, at)$$

$$\int te^{at} \, dt = \frac{1}{a^2}e^{at}(at - 1)$$

$$\int t^2 e^{at} \, dt = \frac{1}{a^3}e^{at}(a^2 t^2 - 2at + 2)$$

$$\int e^{at} \cos bt \, dt = \frac{1}{a^2 + b^2} e^{at}(a \cos bt + b \sin bt)$$

$$\int e^{at} \sin bt \, dt = \frac{1}{a^2 + b^2} e^{at}(a \sin bt - b \cos bt)$$

$$\int \frac{1}{x^2 + a^2} \, dx = \frac{1}{a} \tan^{-1} \frac{x}{a}$$

Signals

$$\mathcal{E}_f = \int_{-\infty}^{\infty} |f(t)|^2 dt \text{ (complex);}$$

$$P_f = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |f(t)|^2 dt;$$

$$\text{rms power} = \sqrt{P_f}$$

Cont; analog; periodic (extension);

(non/anti)causal; energy/power (both);

deterministic/stochastic (info)

$$\int f(t) \cdot \delta(t - t_0) dt = f(t_0) \text{ (} f \text{ cont at } t_0 \text{)}$$

$$f(2x - 6): \text{ shift by 6, scale by 2;}$$

$$f(2(x - 6)): \text{ scale by 2, shift by 6}$$

$$f_e(t) = \frac{1}{2}[f(t) + f(-t)]$$

$$f_o(t) = \frac{1}{2}[f(t) - f(-t)]$$

Systems

$$\mathcal{T}: \sum_{k=0}^{\infty} a_k D^k y(t) = \sum_{l=0}^{\infty} b_l D^l f(t)$$

$$\text{Linear } \mathcal{T}[kf_1(t) + f_2(t)] = ky_1(t) + y_2(t).$$

$$\text{Lin if } a_k, b_l \text{ are not func of } y(t), f(t)$$

$$\text{E. } \sin \dot{y}(t) + t^2 y(t) = (t + 3)f(t)$$

$$\text{Time-inv } \mathcal{T}[f(t - \tau)] = y(t - \tau).$$

$$a_k, b_l \text{ indep of } t \text{ (const coeff)}$$

$$\text{Let } g(t) \equiv f(t - \tau), \text{ find } z(t) = \mathcal{T}[g(t)],$$

$$\text{cmp } y(t - \tau)$$

$$\text{Causal } y(t) \text{ dep only on } f(\tau), \tau \leq t.$$

Compare t and τ .

Ins/dyn y only dep f at present (no $\int$, no memory)

Invertible given $y(t)$, we can know $f(t)$

Conv prop

$$c(t) \equiv \int_{-\infty}^{\infty} f(\tau)g(t - \tau) \, d\tau$$

$$c[n] \equiv \sum_{m=-\infty}^{\infty} f[m]g[n - m]$$

$$f * g = g * f$$

$$f * (g + h) = f * h + g * h$$

$$f * (g * h) = (f * g) * h$$

$$\text{pf: } f * (g * h) = f * (h * g)$$

$$= \int f(\tau_1) \int h(\tau_2) g(t - \tau_2 - \tau_1) \, d\tau_2 \, d\tau_1$$

$$= \int h(\tau_2) \int f(\tau_1) g(t - \tau_1 - \tau_2) \, d\tau_1 \, d\tau_2$$

$$f(t - T_1) * g(t - T_2) = c(t - T_1 - T_2)$$

$$f(at) * g(at) = \left|\frac{1}{a}\right| c(at) \text{ (even/odd)}$$

$$f^{(m)}(t) * g^{(n)}(t) = c^{(m+n)}(t)$$

$$\text{pf: } \dot{f}(\tau) = \lim_{T \rightarrow 0} f(\tau) - f(\tau - T)$$

Graph: shift **left** by $+t$, and reflect;

Every τ replaced by $t - \tau$; Reverted

Conv table

$$f(t) * \delta(t - T) = f(t - T)$$

$$u(t) * u(t) = t u(t)$$

$$e^{at} u(t) * u(t) = \frac{1 - e^{at}}{-a} u(t)$$

$$e^{at} u(t) * e^{bt} u(t) = \frac{e^{at} - e^{bt}}{a - b} u(t)$$

$$a = b, te^{at} u(t)$$

$$e^{at} u(t) * e^{bt} u(-t) = \frac{e^{at} u(t) + e^{bt} u(-t)}{b - a}$$

$$te^{at} u(t) * e^{at} u(t) = \frac{1}{2} t^2 e^{at} u(t)$$

$$t^m u(t) * t^n u(t) = \frac{m! n!}{(m+n+1)!} t^{m+n+1} u(t)$$

Don't forget $[u(t + T_1) - u(t - T_2)]$ term

LTI response

$$Q(D)y(t) = P(D)f(t), \text{ typically}$$

integrating f

Assume causal input $f(t)u(t)$

$$y_{zs}(t) = f(t) * h(t) \text{ from input}$$

$$y_{zs}(0^-) = 0, y_{zs}(0^+) \neq 0$$

Let $h(t) = \mathcal{T}[\delta(t)]$ (impulse response)

$$y_{zs}(t) = \mathcal{T}[f(t)] = \mathcal{T}[f(t) * \delta(t)]$$

$$= \mathcal{T}[\lim \sum f(n\Delta\tau)\delta(t - n\Delta\tau)\Delta\tau]$$

$$= \lim \sum f(n\Delta\tau)h(t - n\Delta\tau)\Delta\tau = f * h$$

$$y_{zi}(t) \text{ from ini, } f(t) = 0, Qy_{zi}(t) = 0$$

$$y_{zi}(0^-) = y_{zi}(0^+), y'_{zi}(0^-) = y'_{zi}(0^+)$$

Ortho set

$$\mathcal{E}_e = \int_{t_1}^{t_2} [e(t)]^2 dt = \int_{t_1}^{t_2} f^2(t) dt$$

$$-2 \sum c_i \int f(t)x_i(t) dt + \int (\sum c_i x_i(t))^2 dt$$

$$= \mathcal{E}_f - 2 \sum \langle f, x_i \rangle + (\sum c_i^2 \int x_i(t)^2 dt +$$

$$\sum_{i \neq j} c_i c_j \int_{t_1}^{t_2} x_i(t)x_j(t) dt)$$

$$\frac{\partial \mathcal{E}_e}{\partial c_i} = 0 = -2 \langle f(t), x_i(t) \rangle + 2\mathcal{E}_i c_i$$

$$\mathcal{E}_e^{\min} = \mathcal{E}_f - \sum_{i=1}^N c_i^2 \mathcal{E}_i$$

$$c_i = \frac{1}{\mathcal{E}_i} \langle f, x_i \rangle = \frac{\int f(t)x_i(t) dt}{\int x_i^2(t) dt}$$

For ortho, $E_z = E_x + E_y$

$$|u + v|^2 = |u|^2 + |v|^2 + u^*v + v^*u$$

$$\langle x(t), y(t) \rangle = \int_{t_1}^{t_2} x(t)y(t)^* dt$$

FS

$$a_0 = \frac{1}{T_0} \int_{T_0} f(t) dt$$

$$a_n = \frac{2}{T_0} \int_{T_0} f(t) \cos(n\omega_0 t) dt$$

$$b_n = \frac{2}{T_0} \int_{T_0} f(t) \sin(n\omega_0 t) dt$$

Energy: T_0 for $n = 0$; $T_0/2$ else

$$\text{Half wave sym } f(t - \frac{T_0}{2}) = -f(t)$$

$$a_{n_{\text{odd}}} = \frac{4}{T_0} \int_0^{T_0/2} f(t) \cos(n\omega_0 t) dt$$

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_0 t}$$

$$F_n = \frac{1}{T_0} \int_{T_0} f(t) e^{-jn\omega_0 t} dt$$

$$C_n \cos(n\omega_0 t + \theta_n) =$$

$$\frac{C_n}{2} (e^{j(n\omega_0 t + \theta_n)} + e^{-j(n\omega_0 t + \theta_n)})$$

$$= (\frac{C_n}{2} e^{j\theta_n}) e^{jn\omega_0 t} + (\frac{C_n}{2} e^{-j\theta_n}) e^{-jn\omega_0 t}$$

$$F_n = \frac{C_n}{2} e^{j\theta_n} = \frac{1}{2}(a_n - jb_n) = |F_n| e^{j\angle F_n}$$

$$F_{-n} = \frac{C_n}{2} e^{-j\theta_n} = \frac{1}{2}(a_n + jb_n)$$

Weak: finite $\int$, fin bounds a, b , fin power

Strong: fin m/m/disc over T_0 , Converge

FS prop

$$\text{Time shift } f(t - t_0) \rightarrow F_n e^{-jn(\omega_0 t_0)}$$

$$|F_n| \text{ same; } \angle F_n \text{ shifted by } -(\omega_0 t_0)n$$

$$\text{Reversal } f(-t) \rightarrow F_{-n}$$

$$\text{Scaling } T = \frac{T_0}{a}, \omega = a\omega_0$$

Multiplication (same T_0):

$$f(t)g(t) \rightarrow F_n * G_n$$

$$\frac{1}{T_0} \int_{T_0} f(t)g(t) e^{jn\omega_0 t} dt$$

$$=$$

$$\frac{1}{T_0} \int (\sum F_m e^{jm\omega_0 t}) (\sum G_k e^{jk\omega_0 t}) e^{-jn\omega_0 t} dt$$

$$= \sum_m \sum_k F_m G_k \frac{1}{T_0} \int_{T_0} e^{j(m+k-n)\omega_0 t} dt$$

$$= \sum_m \sum_k F_m G_k \langle e^{j(m+k)\omega_0 t}, e^{jn\omega_0 t} \rangle$$

$$= \sum_{k=-\infty}^{\infty} G_k F_{n-k}$$

$$\text{Conjugation } f(t)^* = F_n^*$$

Parseval (power sig):

$$P_f = \frac{1}{T_0} \int_{T_0} f(t)f(t)^* dt$$

$$= \frac{1}{T_0} \int_{T_0} (\sum_n F_n e^{jn\omega_0 t}) (\sum_m F_m e^{jm\omega_0 t})^* dt$$

$$= \sum_n \sum_m F_n F_m^* \frac{1}{T_0} \int_{T_0} e^{j(n-m)\omega_0 t} dt$$

$$= \sum_n |F_n|^2 \cdot 1$$

f real $\rightarrow |F|$ even, $\angle F$ odd

f real, even $\rightarrow F$ re, e; $F_{-n} = F_n = F_n^*$

f re, od $\rightarrow F$ im, o; $-F_{-n} = F_n = -F_n^*$

$$f_e(t) \rightarrow \Re\{F_n\}$$

$$f_o(t) \rightarrow j \Im\{F_n\}$$

FT

$$\text{Let } F(\omega) \equiv \int f(t)e^{-j\omega t} dt$$

$$F_n = \frac{1}{T_0} \int_{T_0} f(t)e^{-jn\omega_0 t} dt$$

Limit as $\omega_0 = \Delta\omega \rightarrow 0$,

$$F_n = \frac{\Delta\omega}{2\pi} \int f(t)e^{-jn\Delta\omega t} dt \equiv \frac{\Delta\omega}{2\pi} F(n\Delta\omega)$$

$$f_{T_0}(t) = \sum F_n e^{jn\omega_0 t} = \sum \frac{\Delta\omega}{2\pi} F(n\Delta\omega) e^{jn\Delta\omega t}$$

$$f(t) = \lim_{T_0 \rightarrow \infty} f_{T_0}(t) = \frac{1}{2\pi} \int F(\omega) e^{j\omega t} d\omega$$

$$F(\omega) = |F(\omega)| e^{j\angle F(\omega)}$$

Real signals: amp and phase symmetry

Existence: energy signal ($|e^{-j\omega t}| = 1$)

Strong: fin num max/min/discont

FT Table

$$\delta(t) \rightarrow 1$$

$$1 \rightarrow 2\pi\delta(\omega)$$

$$e^{j\omega_0 t} \rightarrow 2\pi\delta(\omega - \omega_0)$$

$$\cos \omega_0 t \rightarrow \pi[\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$$

$$\sin \omega_0 t \rightarrow j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$$

$$\sum \delta(t - nT_0) \rightarrow \omega_0 \sum \delta(\omega - n\omega_0) \quad \omega_0 = \frac{2\pi}{T_0}$$

$$e^{-at} u(t) \rightarrow \frac{1}{a+j\omega} \quad a > 0$$

$$e^{-a|t|} \rightarrow \frac{2a}{a^2 + \omega^2} \quad a > 0$$

$$u(t) = \lim_{a \rightarrow 0} e^{-at} u(t) \rightarrow \lim_{a \rightarrow 0} \frac{1}{a+j\omega} = \lim_{a \rightarrow 0} \left(\frac{a}{a^2 + \omega^2} - j \frac{\omega}{a^2 + \omega^2} \right) = \pi\delta(\omega) + \frac{1}{j\omega}$$

$$\text{sgn}(t) \rightarrow \frac{2}{j\omega}$$

$$t^n e^{-at} u(t) \rightarrow \frac{n!}{(a+j\omega)^{n+1}} \quad a > 0$$

$$c\omega_0 t u(t) \rightarrow \frac{\pi}{2} (\delta(\omega - \omega_0) + \delta(\omega + \omega_0)) + \frac{j\omega}{\omega_0^2 - \omega^2}$$

$$s\omega_0 t u(t) \rightarrow \frac{\pi}{2j} (\delta(-) - \delta(+)) + \frac{\omega_0}{\omega_0^2 - \omega^2}$$

$$e^{-at} \cos \omega_0 t u(t) \rightarrow \frac{a+j\omega}{(a+j\omega)^2 + \omega_0^2} \quad a > 0$$

$$e^{-at} \sin \omega_0 t u(t) \rightarrow \frac{\omega_0}{(a+j\omega)^2 + \omega_0^2} \quad a > 0$$

$$\text{rect}\left(\frac{t}{\tau}\right) \rightarrow \tau \text{sinc}\left(\frac{\tau}{2}\omega\right)$$

$$\frac{W}{\pi} \text{sinc}(Wt) \rightarrow \text{rect}\left(\frac{\omega}{2W}\right)$$

$$\Delta\left(\frac{t}{\tau}\right) \rightarrow \frac{\tau}{2} \text{sinc}^2\left(\frac{\tau}{4}\omega\right)$$

$$\frac{W}{2\pi} \text{sinc}^2\left(\frac{W}{2}t\right) \rightarrow \Delta\left(\frac{\omega}{2W}\right)$$

$$\begin{aligned} [\omega^2 \text{r}\left(\frac{\omega}{2\omega_0}\right)] &\leftarrow \frac{1}{2\pi} \frac{e^{j\omega t}}{(jt)^3} (-\omega^2 t^2 - 2j\omega t + 2)_{-\omega_0} \\ &= \frac{(\omega_0^2 t^2 - 2) \sin \omega_0 t + 2\omega_0 t \cos \omega_0 t}{\pi t^3} \end{aligned}$$

$$\left[\frac{|\omega|}{\omega_0} \text{rect}\left(\frac{\omega}{2\omega_0}\right)\right] \leftarrow \frac{\cos \omega_0 t + \omega_0 t \sin \omega_0 t - 1}{\omega_0 \pi t^2}$$

Frequency domain prop

Linearity

Time shift $f(t-t_0) \rightarrow F(\omega)e^{-jt_0\omega}$

$|F|$ unchanged; $\angle F = -t_0\omega$, lin shift

Freq shift $f(t)e^{j\omega_0 t} \rightarrow F(\omega - \omega_0)$

Duality $f(t) \rightarrow F(\omega)$, $F(t) \rightarrow 2\pi f(-\omega)$

$$\begin{aligned} \text{pf. } f(t) &= \frac{1}{2\pi} \int F(\lambda) e^{jt\lambda} d\lambda \\ 2\pi f(-t) &= \int F(\lambda) e^{-tj\lambda} d\lambda = \mathcal{F}[F(\lambda)] \end{aligned}$$

Reversal $f(-t) \rightarrow F(-\omega)$

Scaling $f(at) \rightarrow \frac{1}{|a|} F\left(\frac{\omega}{a}\right)$

Convolution $f * g \rightarrow FG$, $fg \rightarrow \frac{1}{2\pi} F * G$

$$\begin{aligned} \mathcal{F}[f * g] &= \int e^{-j\omega t} \int f(\tau)g(t-\tau) d\tau dt \\ &= \int f(\tau) \mathcal{F}[g(t-\tau)] d\tau \\ &= \int f(\tau) G(\omega) e^{-j\omega\tau} d\tau \\ &= \frac{1}{2\pi} \mathcal{F}^{-1}[F * G] \\ &= \left(\frac{1}{2\pi}\right)^2 \int e^{j\omega t} \int F(\lambda) G(\omega - \lambda) d\lambda d\omega \end{aligned}$$

Diff $f^{(n)}(t) \rightarrow (j\omega)^n F(\omega)$ (diff $e^{j\omega t}$)

Int $\int_{-\infty}^t f(\tau) d\tau \rightarrow \frac{1}{j\omega} F(\omega) + \pi F(0)\delta(\omega)$

$$\begin{aligned} \int &= f(t) * u(t) \rightarrow F(\omega)U(\omega) \\ U(\omega) &= \lim_{a \rightarrow j\omega} \frac{1}{a+j\omega} = \lim_{a \rightarrow j\omega} \left(\frac{a}{a^2 + \omega^2} - j \frac{\omega}{a^2 + \omega^2} \right) \\ &= \pi\delta(\omega) + \frac{1}{j\omega} \end{aligned}$$

$$\left(\int \frac{a}{\omega^2 + a^2} d\omega = \tan^{-1} = \pi \right)$$

Conjugation $f(t)^* \rightarrow F(-\omega)^*$

Symmetry Re $\rightarrow$ mag even, phase odd

$$\begin{aligned} (F(-\omega) &= F(\omega)^*) \\ \text{real, even} &\rightarrow \text{real, even;} \\ \text{real, odd} &\rightarrow \text{imaginary, odd} \end{aligned}$$

f even: $F(\omega) = 2 \int_0^\infty f(t) \cos(\omega t) dt$

f odd: $F(\omega) = -2j \int_0^\infty f(t) \sin(\omega t) dt$

Parseval

$E_f = \int |f(t)|^2 dt = \frac{1}{2\pi} \int |F(\omega)|^2 d\omega$ for energy sig

$$\begin{aligned} &= \int f f^* dt = \int f(t) \mathcal{F}^{-1}[F(-\omega)^*] dt \\ &= \int f(t) \frac{1}{2\pi} \int F(-\omega)^* e^{j\omega t} d\omega dt \\ &= \frac{1}{2\pi} \int f(t) \int F(\lambda)^* e^{-jt\lambda} d\lambda dt \\ &= \frac{1}{2\pi} \int F(\lambda)^* \int f(t) e^{-j\lambda t} dt d\lambda \end{aligned}$$

$$\Delta E_f = \frac{2}{2\pi} \int_{\omega_1}^{\omega_2} |F(\omega)|^2 d\omega$$

Autocorrelation

$$\psi_f(t) \equiv \int f(\tau) f(\tau - t) d\tau \rightarrow |F(\omega)|^2$$

Modulation

$$m(t) \cos(\omega_c t) \rightarrow \frac{1}{2} [M(\omega + \omega_c) + M(\omega - \omega_c)]$$

$$e(t) = m(t) \cos^2 \omega_c t$$

$$E(\omega) = \frac{1}{2} M + \frac{1}{4} [M(\omega + 2\omega_c) + M(\omega - 2\omega_c)]$$

SSB 1/4 gain

$$\phi_{AM}(t) = [A + f(t)] \cos(\omega_0 t)$$

$$A \geq f(t) \text{ for all } t$$

modulation index $\mu \equiv f_{\max}/A$

$\mu = \infty$, suppressed carrier; $\mu = 1$, marginal

LTIC sys trans, (marginally) stable

Let $e^{j\omega t} \Rightarrow H(\omega)e^{j\omega t}$

$$\begin{aligned} \lim \sum \frac{F(n\Delta\omega)\Delta\omega}{2\pi} e^{jn\Delta\omega t} &\Rightarrow \\ \lim \sum \frac{F(n\Delta\omega)H(n\Delta\omega)\Delta\omega}{2\pi} e^{jn\Delta\omega t} & \\ y(t) &= \frac{1}{2\pi} \int F(\omega)H(\omega) e^{j\omega t} d\omega \\ Y(\omega) &= F(\omega)H(\omega) \end{aligned}$$

Distortionless $y(t) = kf(t - t_d)$, so

$$H(\omega) = k e^{-j\omega t_d}$$

Payley-Wiener H realizable, h causal iff

$$\int \frac{|\ln|H(\omega)||}{1+\omega^2} d\omega < \infty \text{ (consecutive 0s)}$$

$$\text{Truncate } \hat{h}(t) = h(t)u(t)$$

Periodic FT

$$f(t) = \sum_n F_n e^{jn\omega_0 t}$$

$$\mathcal{F}[f(t)] = 2\pi \sum_n F_n \delta(\omega - n\omega_0)$$

$$Y = F(\omega)H(\omega) =$$

$$2\pi \sum F_n H(n\omega_0) \delta(\omega - n\omega_0)$$

$$Y_n \equiv F_n H(n\omega_0). \text{ Periodic with same } \omega_0$$

Eigen: $f(t) = e^{j\omega_0 t}$, $Y_1 = H(1\omega_0)$,

$$y(t) = H(1\omega_0) e^{j\omega_0 t}$$

$f(t) = \cos(\omega_0 t + \theta)$, assume $h(t)$ real

$$\begin{aligned} y &= \frac{1}{2} (e^{j(\theta + \omega_0 t)} H(\omega_0) + e^{-j(\theta + \omega_0 t)} H(-\omega_0)) \\ &= |H(\omega_0)| \cos(\omega_0 t + \theta + \angle H(\omega_0)) \end{aligned}$$

$$\begin{aligned} \cos 2t * e^{-3t} u(t) &\equiv f * h \\ &= |H(2)| \cos(2t + \angle H(2)) \end{aligned}$$

Sampling

$$\bar{f}(t) \equiv f(t) \delta_{T_s}(t) = \sum f(nT_s) \delta(t - nT_s)$$

$$\bar{F}(\omega) = \frac{1}{2\pi} F(\omega) * \left[\frac{2\pi}{T_s} \sum \delta(\omega - n\omega_s) \right]$$

$$= \frac{1}{T_s} \sum F(\omega - n\omega_s)$$

$$\omega_s \geq 4\pi B, F_s \geq F_N \equiv 2B$$

Intrapolation when $F_s = 2B$

$$F(\omega) = \bar{F}(\omega) T_s \text{rect}\left(\frac{\omega}{4\pi B}\right)$$

If $F_s = 2B$, $f(t) = \bar{f}(t) * \frac{2B}{F_s} \text{sinc}(2\pi Bt)$

$$= \sum_n f(nT_s) \delta(t - nT_s) * \text{sinc}(2\pi Bt)$$

$$= \sum_n f(nT_s) \text{sinc}(2\pi Bt - n\pi)$$

ana FS, with basis sinc, interpolation

formula

If $F_s > 2B$, $f(t) = \sum f(nT_s) w(t - nT_s)$

not sinc weight

for some relaxed LP filter $w(t)$

Anti-alias before sampling: LPF of $F_s/2$

Practical sampling

$$p_T(t) = \frac{T}{T_s} + \sum \left(\frac{2}{\pi n} \sin(n\pi \frac{T}{T_s}) \right) \cos(n\omega_s t)$$

$$\begin{aligned} P_T(\omega) &= 2\pi \frac{T}{T_s} \delta(\omega) \\ &+ \sum \frac{\pi 2 \sin(\dots)}{\pi n} [\delta(\omega + n\omega_s) + \delta(\omega - n\omega_s)] \end{aligned}$$

$$\text{Square } \frac{4}{\pi} (\cos t - \frac{1}{3} \cos 3t + \frac{1}{5} \cos 5t - \dots)$$

$$\text{Triangle } \frac{8}{\pi^2} (\sin t - \frac{1}{9} \sin 3t + \frac{1}{25} \sin 5t - \dots)$$

$$\text{Sawtooth } \frac{2}{\pi} (\sin t - \frac{1}{2} \sin 2t + \frac{1}{3} \sin 3t - \dots)$$

$$\delta \text{ train } \delta_{T_0}(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_0)$$